

California House Price Prediction

Machine Learning & Pattern Recognition Project

AI Models | Random Forest vs XGBoost | Data Analysis

Project Team Details

Course

Machine Learning and Pattern Recognition

Group Name

California Housing

Department

Software Engineering

Team Members

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Project Overview

Goal

Predict **Median House Values** in California neighborhoods

Approach

Supervised Learning

Regression Task

Challenge

Non-linear Data

Diverse Features

Workflow

1. Data Exploration
2. Preprocessing
3. Feature Engineering
4. Model Selection
5. Training
6. Evaluation

Comparing Random Forest vs XGBoost performance

Dataset & Tools

California Housing Dataset

- Median Income
- House Age
- Avg Rooms
- Avg Bedrooms
- Population
- Avg Occupancy
- Latitude
- Longitude

Target Variable

Median House Value

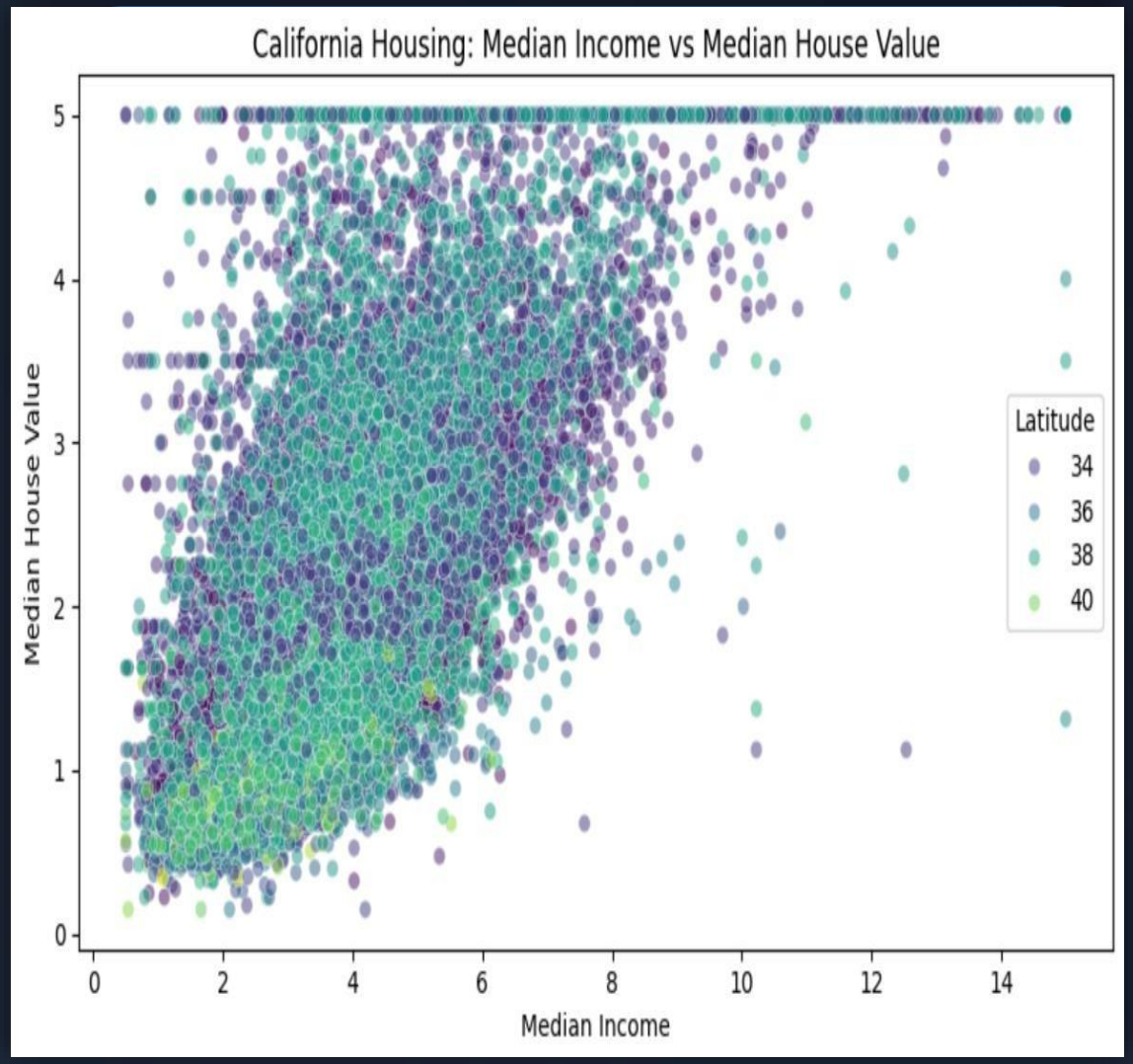


Python Tech Stack

- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Matplotlib
- Seaborn



Data Exploration & Preprocessing



Key Insights

- Coastal areas and big cities are **more expensive**
- High income correlates with **high prices**

Preprocessing Steps

1. Missing Values
2. Feature Selection
3. Feature Scaling
4. Train-Test Split

80/20 Split • **StandardScaler** Normalization • dropna() for missing values

Feature Engineering & Correlation

Engineered Features

StandardScaler Applied

Zero mean, unit variance normalization

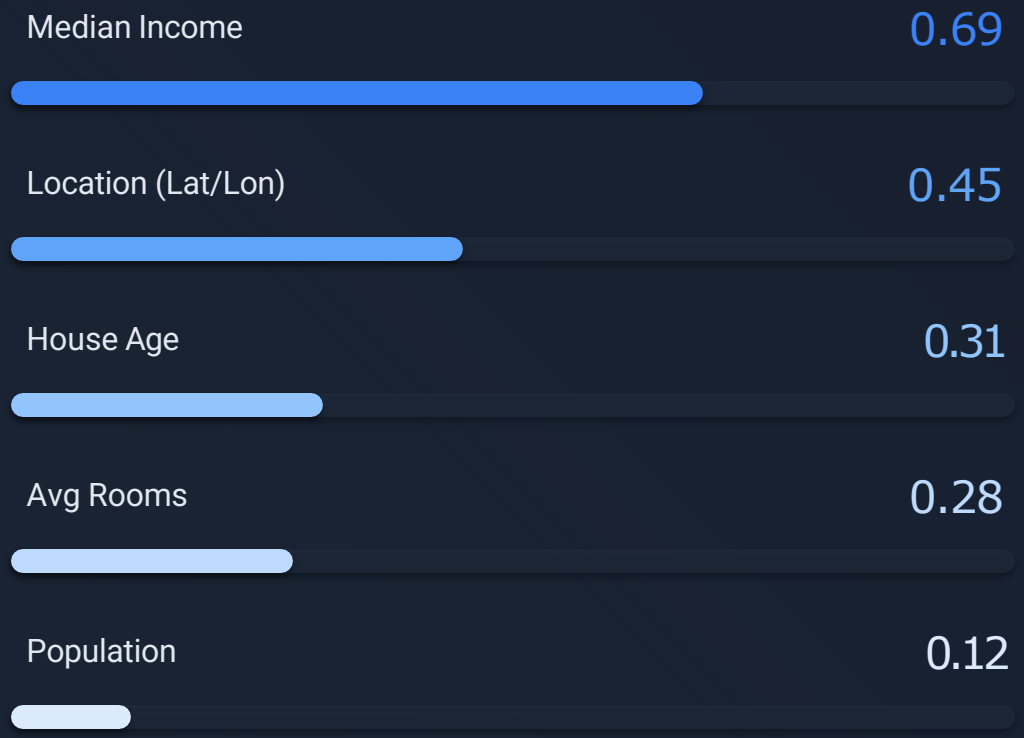
Train/Test Split (80/20)

random_state=42 for reproducibility

Missing Value Handling

dropna() — rows with NaN removed

Correlation Analysis



Median Income is the strongest predictor of house values

Random Forest Model

Bagging Method

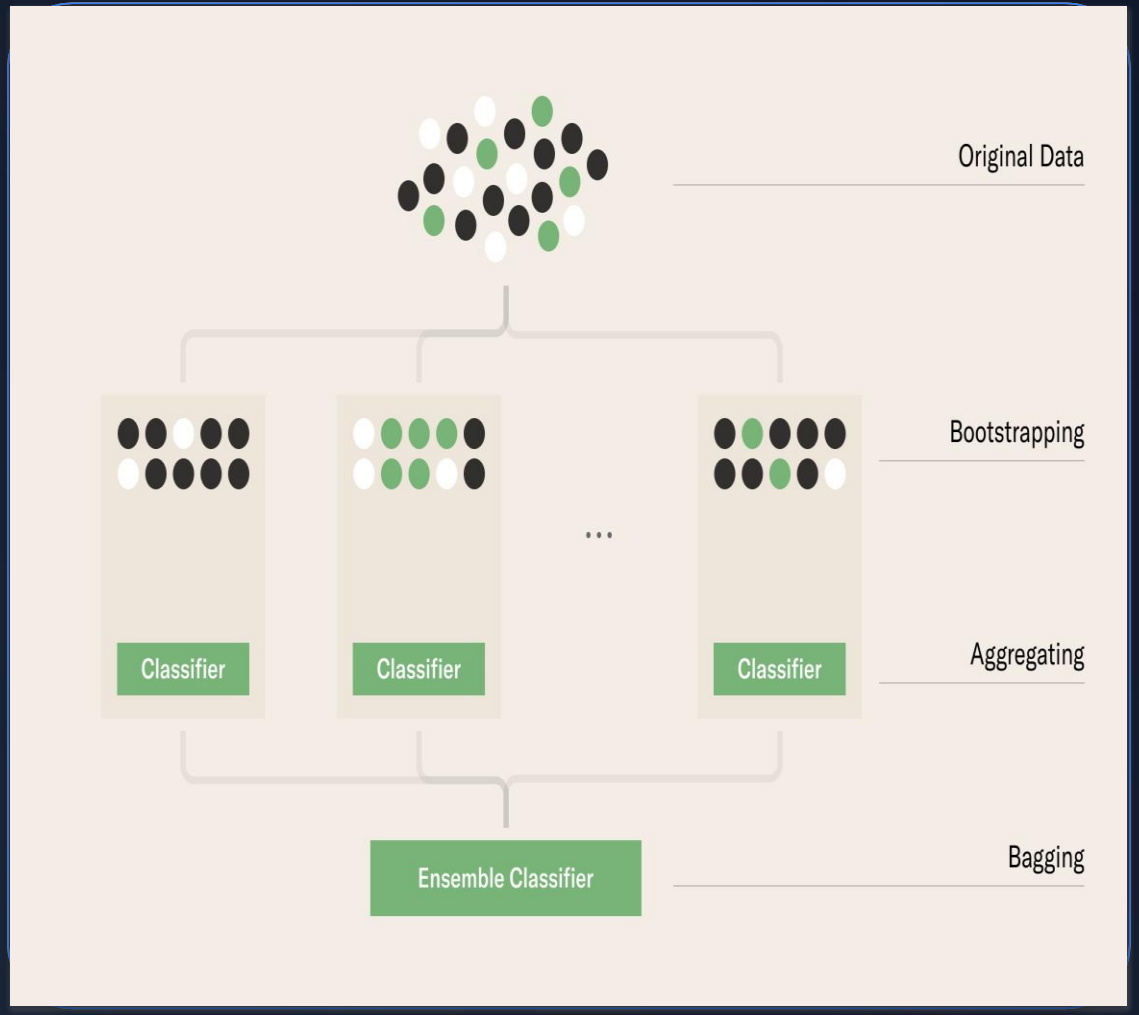
Builds **multiple decision trees** on random subsets and averages predictions

Advantages

- Robust to outliers
- Reduces overfitting
- No scaling required
- Parallel training

Disadvantages

- Computationally expensive
- Less interpretable
- Overfits noisy data
- Memory intensive



XGBoost Model

Extreme Gradient Boosting

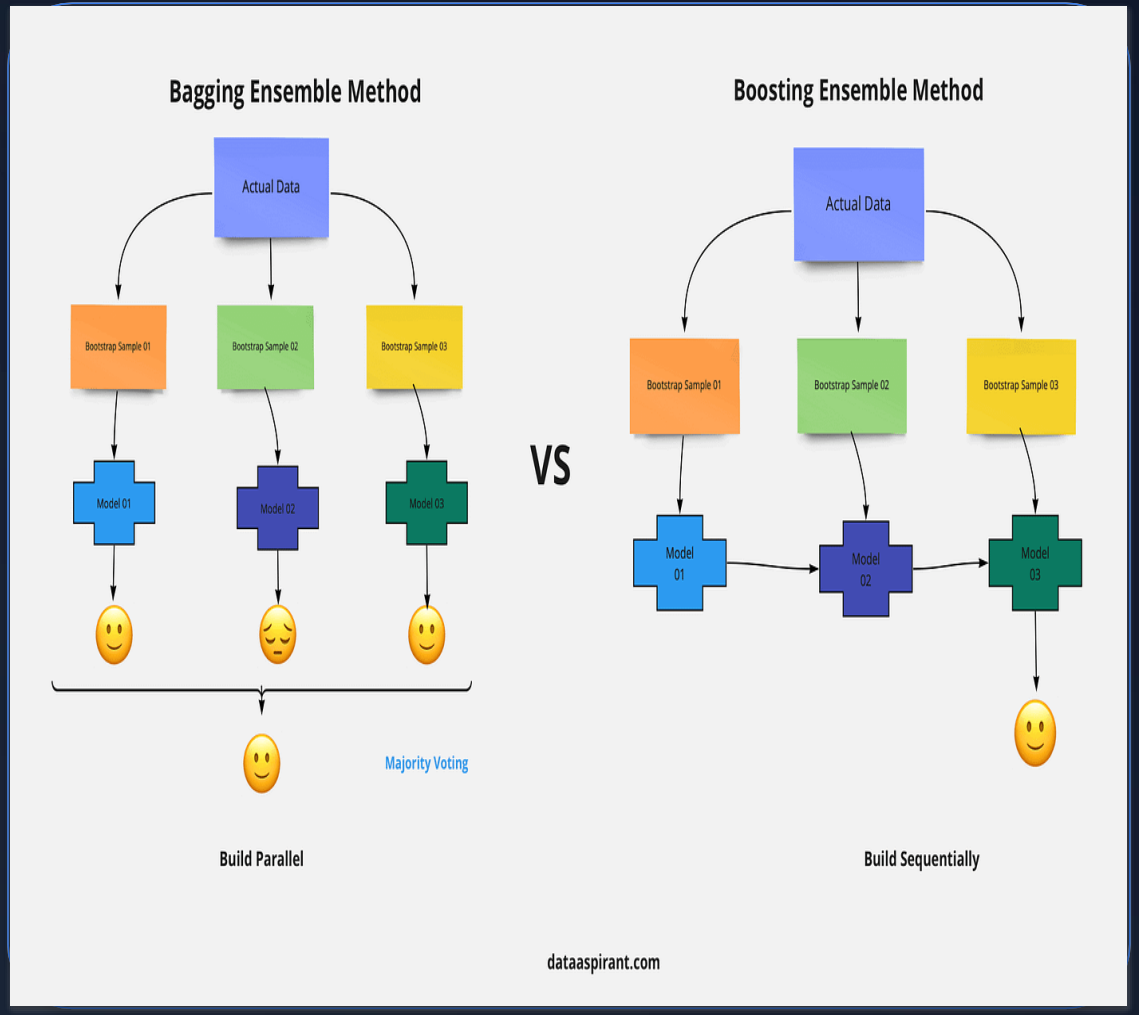
Sequential ensemble that corrects errors of previous trees using L1/L2 regularization

Advantages

- High accuracy
- Fast execution
- Handles missing data
- Built-in regularization

Disadvantages

- Overfitting risk
- Complex tuning
- Sensitive to outliers
- Black box nature



Training Process & Evaluation

Training Steps

1. **Train-Test Split**

   80% Training / 20% Testing

2. **Cross-Validation**

   2-Fold CV on training set

3. **Hyperparameter Tuning**

   RandomizedSearchCV optimization

4. **Model Fitting**

   Train on full training data

Evaluation Metrics

RMSE

Root Mean Squared Error - **Penalizes large errors**, measures prediction accuracy

MAE

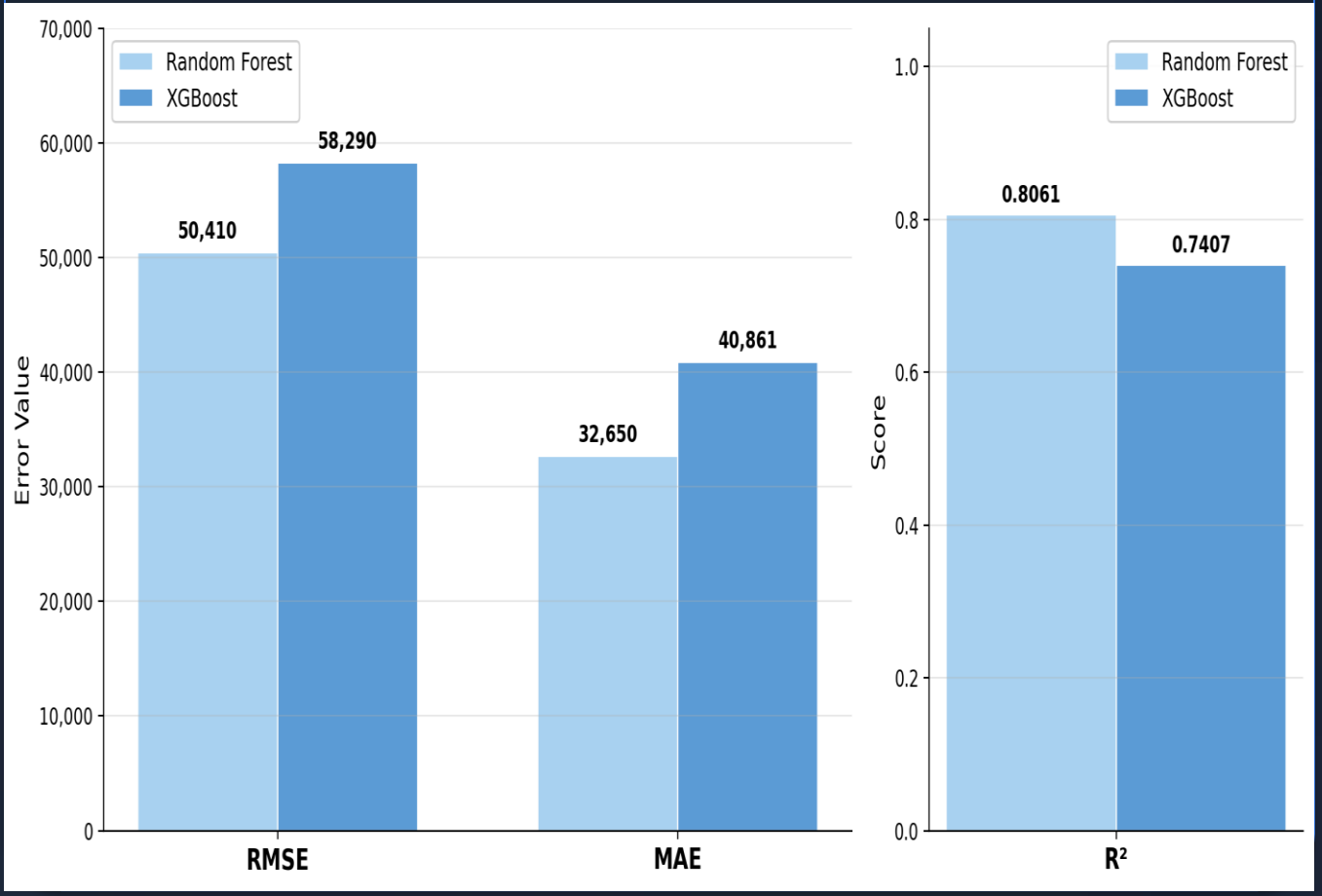
Mean Absolute Error - **Average magnitude** of errors, easier to interpret

R^2 Score

R-Squared - **Proportion of variance explained** (0 to 1)

Results Comparison

Model Performance Comparison



Detailed Metrics

RMSE

Random Forest	XGBoost
$50,410	$58,290

MAE

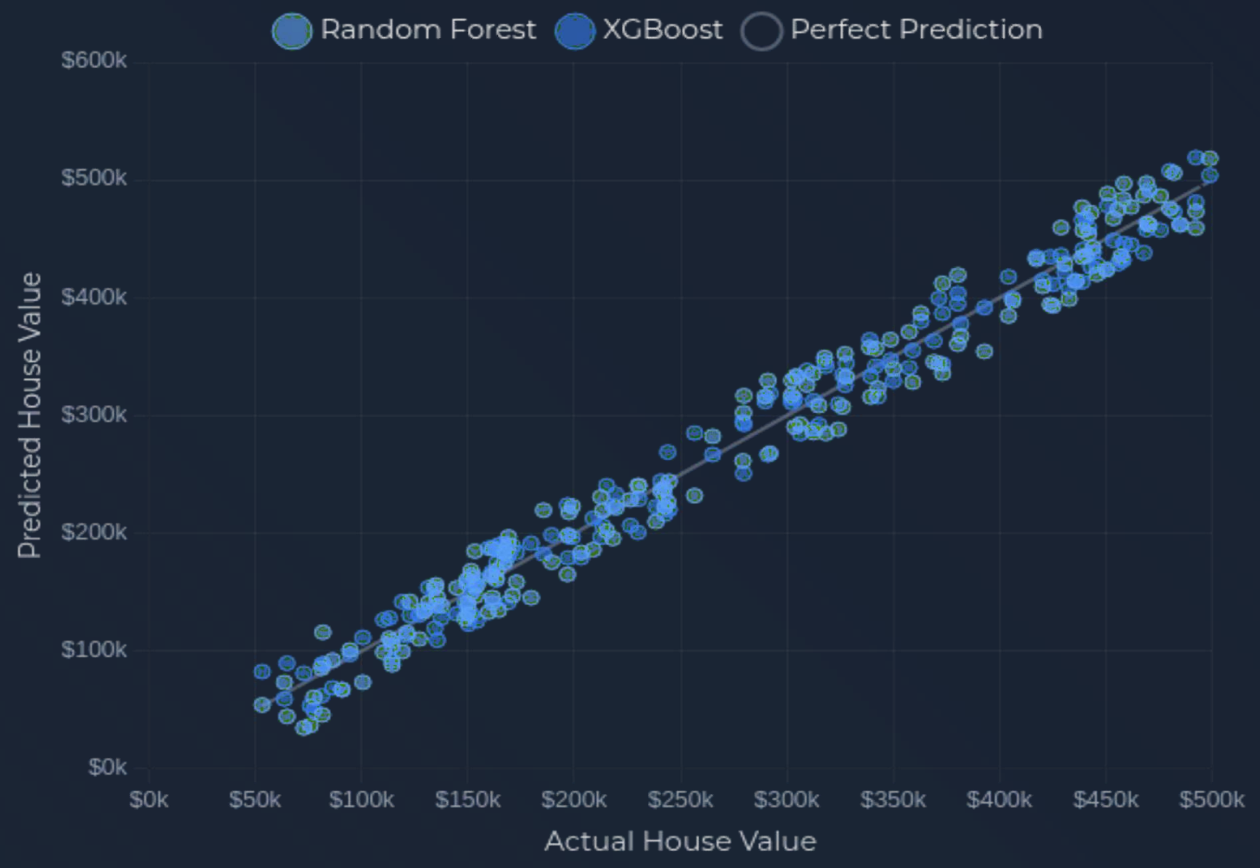
Random Forest	XGBoost
$32,650	$40,861

R^2 Score

Random Forest	XGBoost
0.8061	0.741

Visualizing Predictions

Actual vs Predicted Values



Key Insights

Perfect Prediction Line

Diagonal represents **zero error**

Random Forest Performance

Tighter clustering around diagonal line

Random Forest Spread

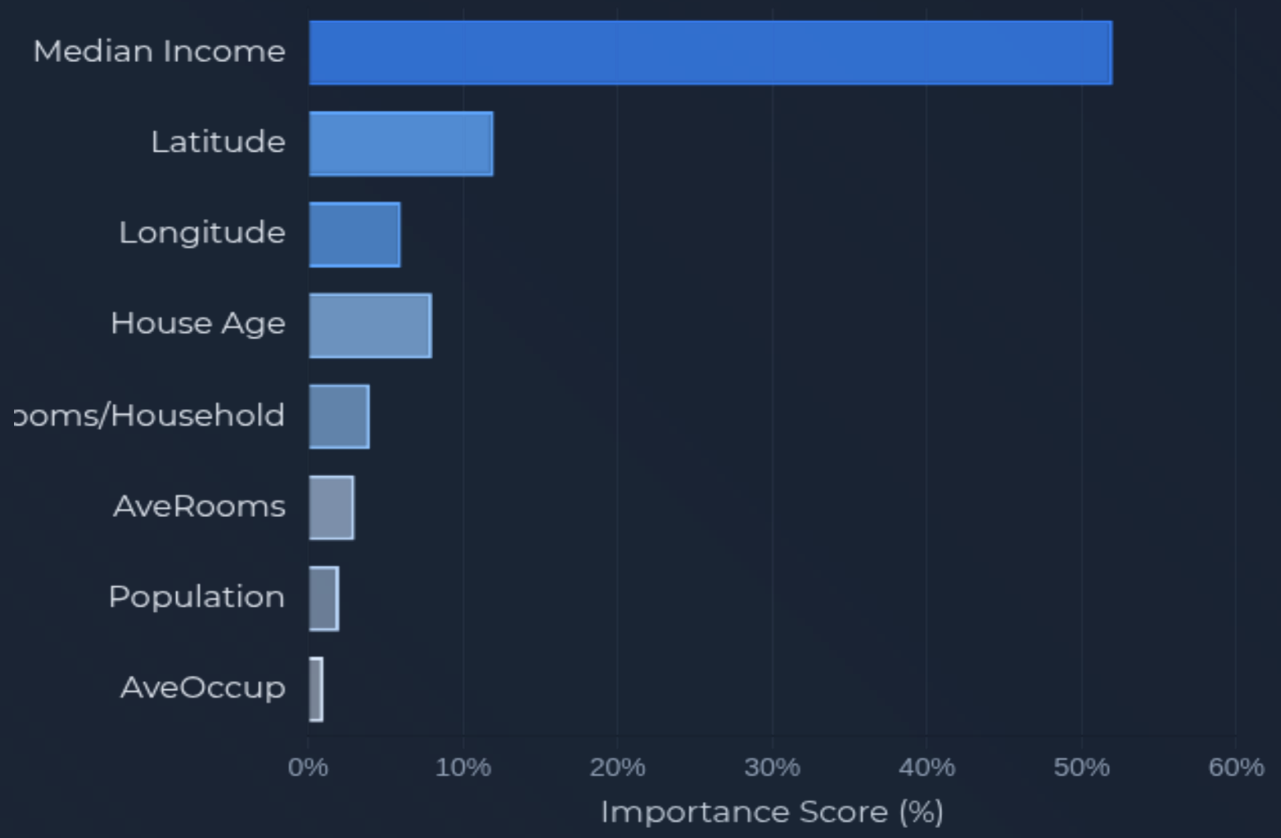
More **variance** in predictions

Winner: Random Forest

Higher accuracy with **lower prediction error**

Feature Importance

Feature Contribution to Price Prediction



Key Predictors

1 **Median Income**

52% importance - Strongest driver

2 **Location (Lat/Lon)**

18% importance - Coastal proximity

3 **House Age**

8% importance - Property maturity

Key Insight

Income and location are the primary factors driving California house prices

Model Comparison Summary

XGBoost

R^2 Score	**0.7407**
RMSE	**$58,290**
MAE	**$40,861**

Advantages

- Easier to implement
- Less hyperparameter tuning
- Robust to outliers

Disadvantages

- Lower accuracy
- Slower training
- Less interpretable

Random Forest

R^2 Score	0.8061
RMSE	$50,410
MAE	$32,650

Advantages

- Higher accuracy
- Fast execution
- Built-in regularization

Disadvantages

- Complex tuning required
- Overfitting risk if not tuned
- Sensitive to outliers

Why We Chose These Models (Random Forest & XGBoost)

We chose Random Forest and XGBoost because house prices depend on many factors in complex, non linear ways for example, how location and income interact. Linear models would struggle with this.

Random Forest gave us a strong, easy to use model with very little tuning. It reached an R^2 score of 0.8061, meaning it explains over 80% of the price differences.

XGBoost is another powerful model that handles the dataset's $500,000 price cap well. It achieved a solid R^2 of 0.7407. Using both let us compare two popular approaches.

In the end, we picked Random Forest as our final model because it was more accurate and simpler to set up.

Conclusion

Random Forest is the Winner

R^2 Score **0.8061**

Higher accuracy in price prediction

RMSE **$50,410**

Lower prediction error vs XGBoost

Best performing model for California house price prediction

Key Findings

- **Income** is the #1 driver of house prices
- **Location** (coastal proximity) is critical
- **House age** and **rooms** also influence prices

Random Forest

Strong baseline model with **easier implementation** and less tuning required

Future Work & Q&A

Future Improvements

Recent Data

Collect data beyond 1990 census for current market trends

External Features

School ratings, crime rates, proximity to amenities

Deep Learning

Explore Neural Networks for complex pattern recognition

Time-Series Analysis

Track price trends over time for market forecasting

Questions & Answers

Thank you for your attention!

We welcome any questions or suggestions.

Contact Information

California Housing Group • Software Engineering Department